

| Question Number | Answer                                                                                                                                     | Additional Guidance                                                                   | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------|
| <b>2(a)</b>     | An answer that includes: <ul style="list-style-type: none"> <li>• two structures named (1)</li> <li>• third structure named (1)</li> </ul> | e.g. chloroplast, amyloplast, mitochondria, Golgi body, rough ER, smooth ER, lysosome | (2)  |

| Question Number | Answer                                                                                                                                                                                           | Additional Guidance                                                                    |                                       | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------|------|
| 2(b)(i)         | A table drawn with the following features: <ul style="list-style-type: none"><li>• suitable table drawn (1)</li><li>• headings with units (1)</li><li>• all data correctly entered (1)</li></ul> | Concentration of chemical / ppm                                                        | Concentration of ions released / a.u. |      |
|                 |                                                                                                                                                                                                  | 15                                                                                     | 5                                     |      |
|                 |                                                                                                                                                                                                  | 25                                                                                     | 40                                    |      |
|                 |                                                                                                                                                                                                  | 50                                                                                     | 80                                    |      |
|                 |                                                                                                                                                                                                  | 75                                                                                     | 140                                   |      |
|                 |                                                                                                                                                                                                  | 100                                                                                    | 160                                   |      |
|                 |                                                                                                                                                                                                  | 125                                                                                    | 230                                   |      |
|                 |                                                                                                                                                                                                  | 150                                                                                    | 250                                   |      |
|                 |                                                                                                                                                                                                  | Allow leeway of 1 either way on concentrations on x<br>extra data (pigment) loses mp 3 |                                       |      |
| (3)             |                                                                                                                                                                                                  |                                                                                        |                                       |      |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                            | Additional Guidance                                                 | Mark       |
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| <b>2(b)(ii)</b> | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• as concentration (of chemical) increases conc. of pigment and ions (released) increase (1)</li> <li>• thus {permeability / described} (to both ions and pigment) increases (1)</li> <li>• permeability to ions less than to pigment (1)</li> <li>• there is no indication of variability in the data (1)</li> </ul> | <p>ACCEPT Positive correlation</p> <p>ACCEPT no SD / error bars</p> | <b>(3)</b> |

| Question Number  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Additional Guidance                                                                                                                                                                                                                                                  | Mark       |
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| <b>2(b)(iii)</b> | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• equal sized pieces of plant tissue cut (1)</li> <li>• washed (in distilled water until no more pigment lost) (1)</li> <li>• placed in a range of <u>7</u> concentrations of the chemical shown (1)</li> <li>• samples of the liquid (around the discs) were removed / tissue removed from tube (1)</li> <li>• method for measurement of intensity of colour described (1)</li> <li>• one variable identified</li> </ul> | <p>Accept alternatives e.g.mass</p> <p>Accept rinsed</p> <p>Allow 8 if it is clear one is distilled water or 0%</p> <p>colorimeter (or implied), looking down tubes, compare with colour standard</p> <p>e.g. temperature, pH, time in water, volume of solution</p> | <b>(5)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                        | Additional Guidance                           | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------|
| <b>2(c)</b>     | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• chemical {damages/destroys} membranes (1)</li> <li>• pigment leaks later than ions / described (1)</li> <li>• pigment in vacuole (only) (1)</li> <li>• it will take more time for tonoplast and cell membrane to be damaged than for just the cell membrane (1)</li> </ul> | <p>Ignore wall</p> <p>Accept in tonoplast</p> | <b>(4)</b> |